Acute Inflammation



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Objectives



At the end of the session students should be able to:

- Define acute inflammation
- Name causes of inflammation
- Enumerate cardinal signs of inflammation
- Describe the vascular changes and cellular events of acute inflammation

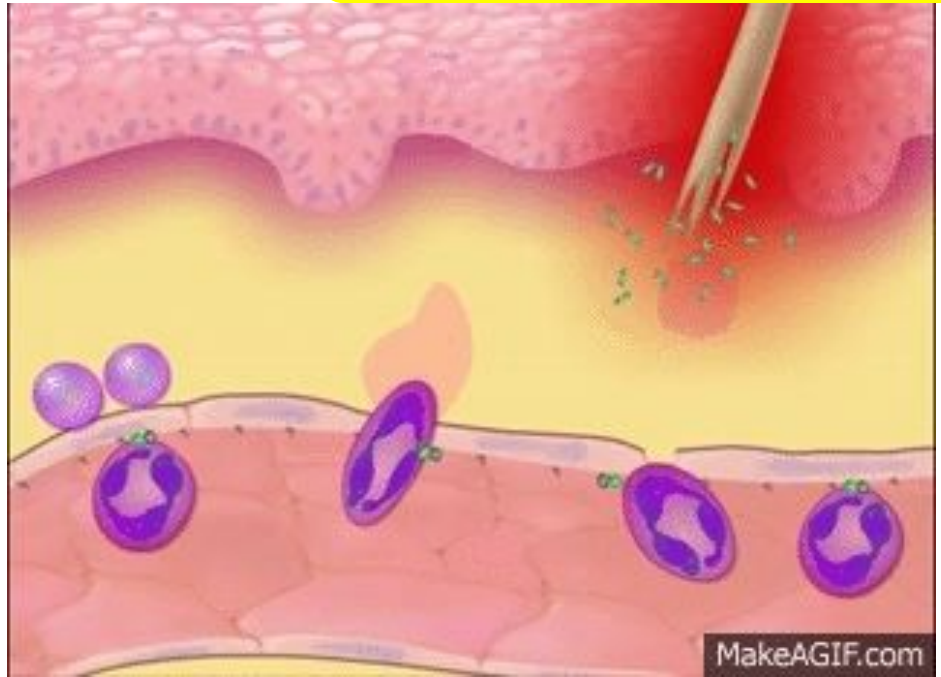


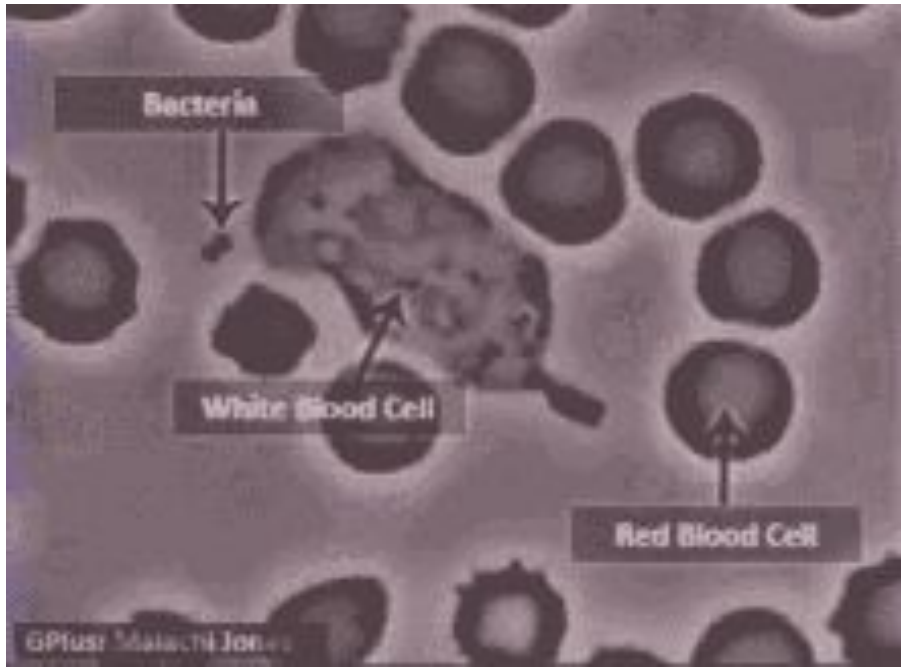
Inflammation??



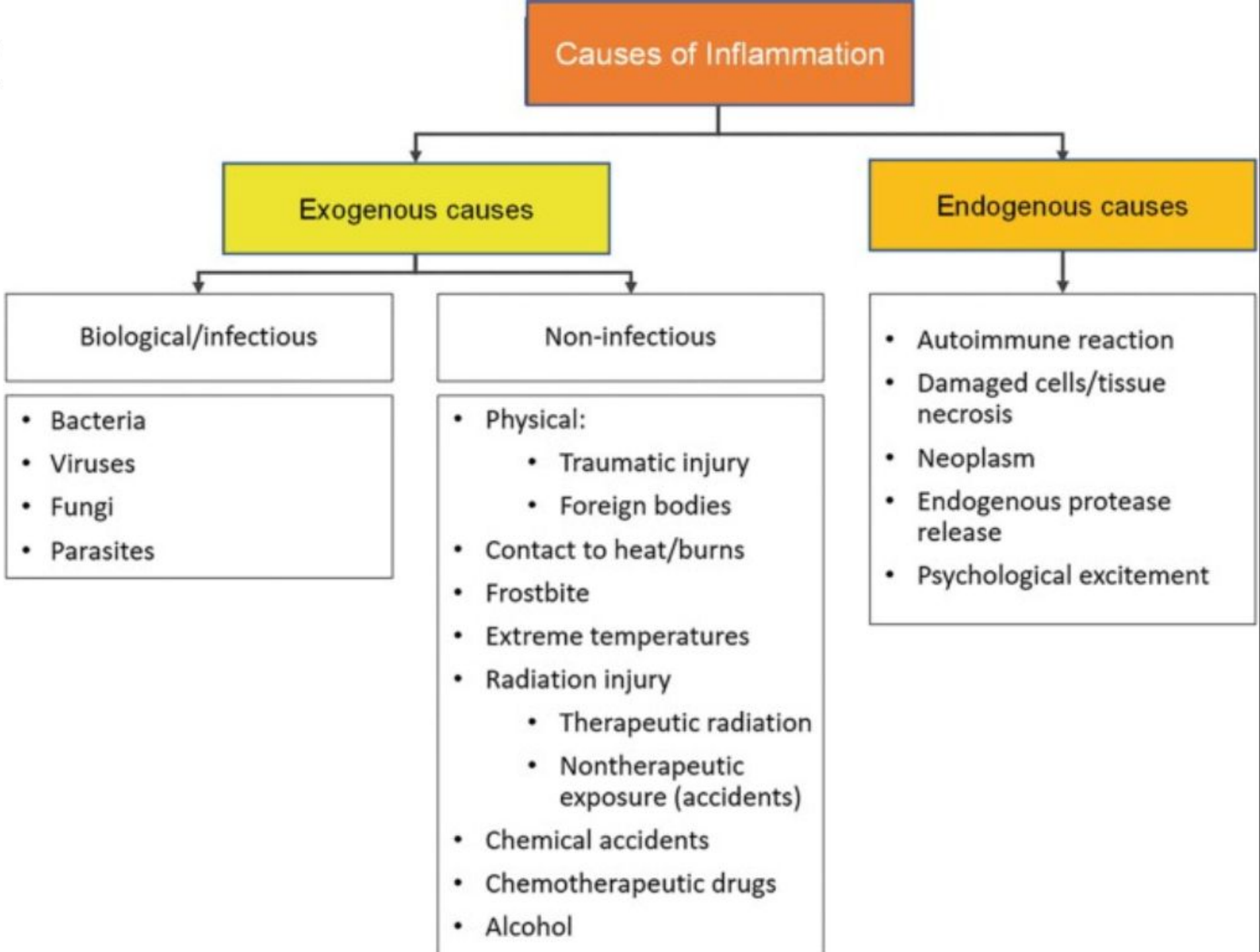
Inflammation

Inflammation is a response of vascularized tissues to infections and damaged tissues that brings cells and molecules of host defense from the circulation to the sites where they are needed, in order to eliminate the offending agents





Inflammation



Cardinal Signs of Inflammation



Injury



Rubor, calor

Vasodilatation



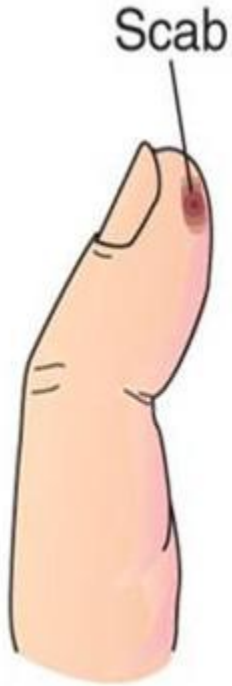
Tumor

Increased vascular permeability



Dolor

Cytokines

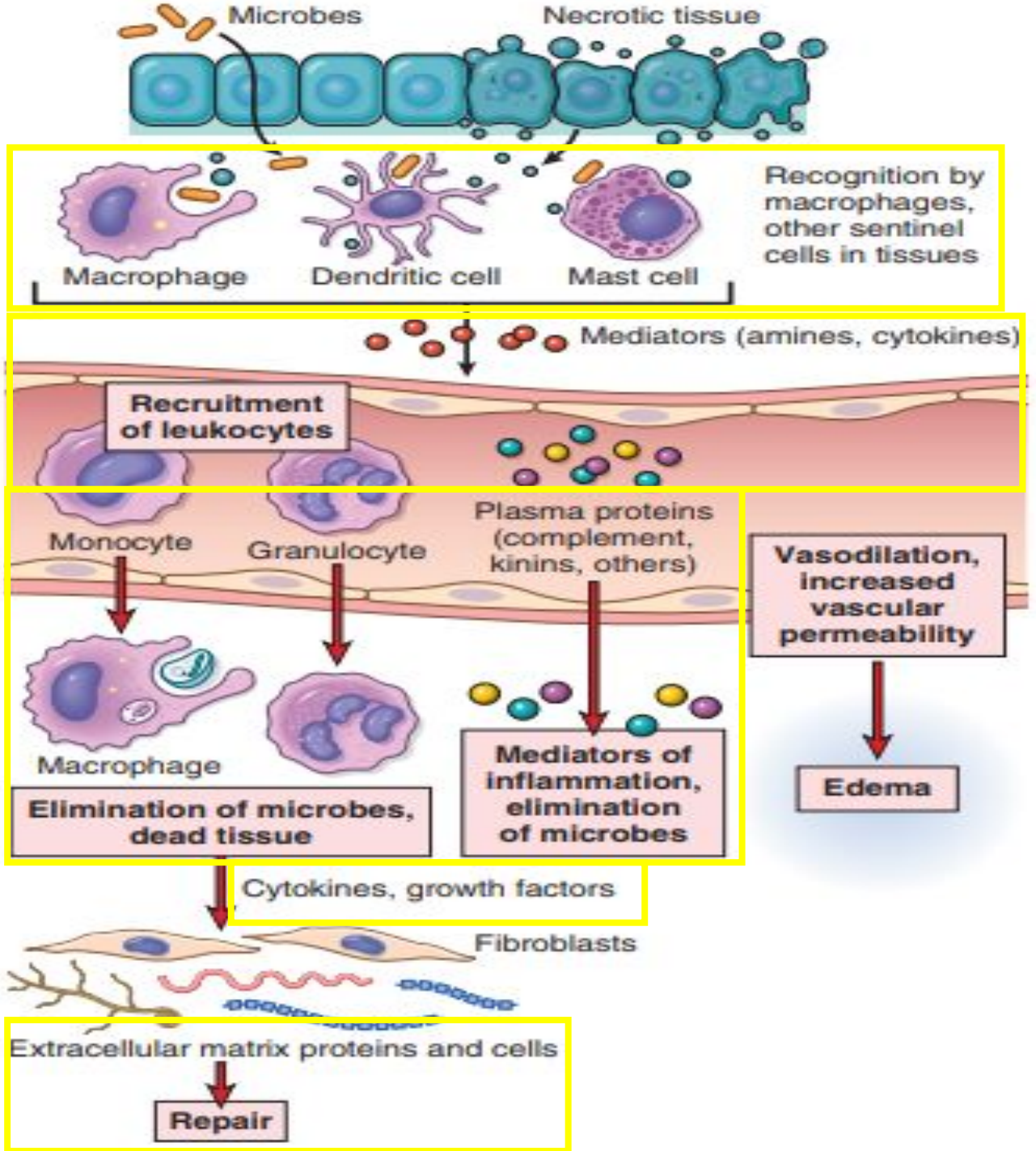


Loss of function

Pain
Tissue damage

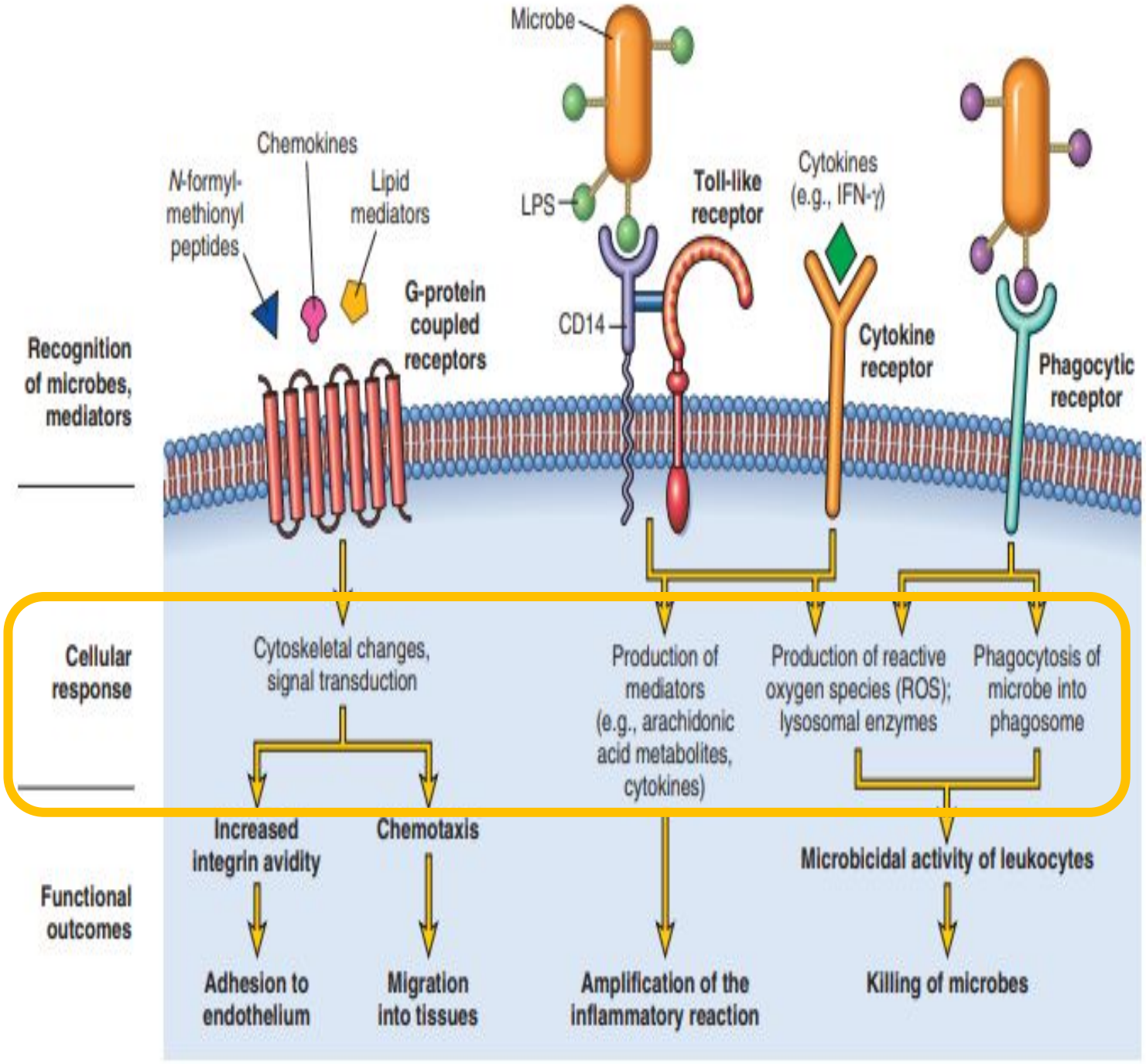
Inflammation

- Recognition of injurious agent
- Recruitment of leukocytes
- Removal of the agent
- Regulation of response
- Resolution



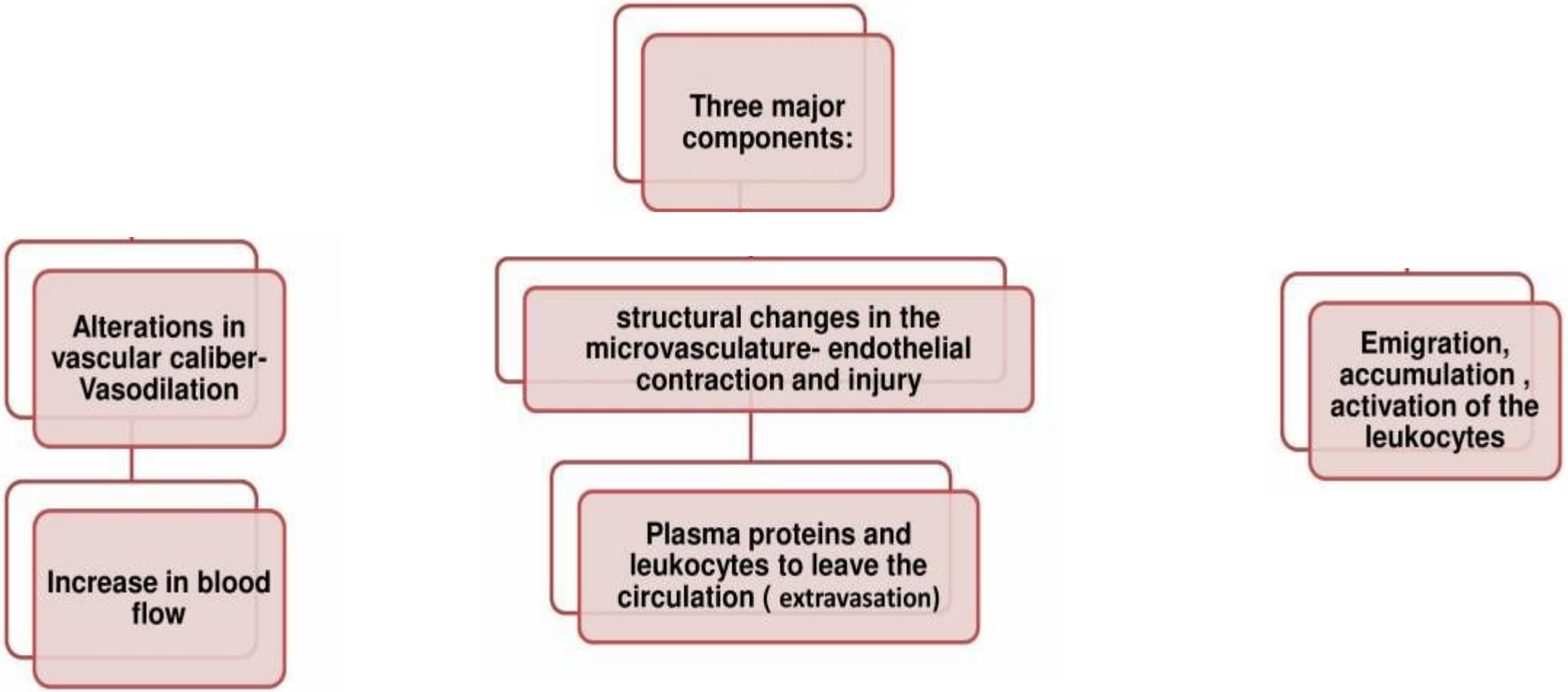
Leukocyte Activation

- Toll-like receptors (TLRs) for foreign invaders
- Cytosolic receptors for recognition of molecules of cell damage
- Receptors for the Fc tails and complement proteins, recognize opsonised microbes
- Mannose-binding lectin recognizes microbial sugars



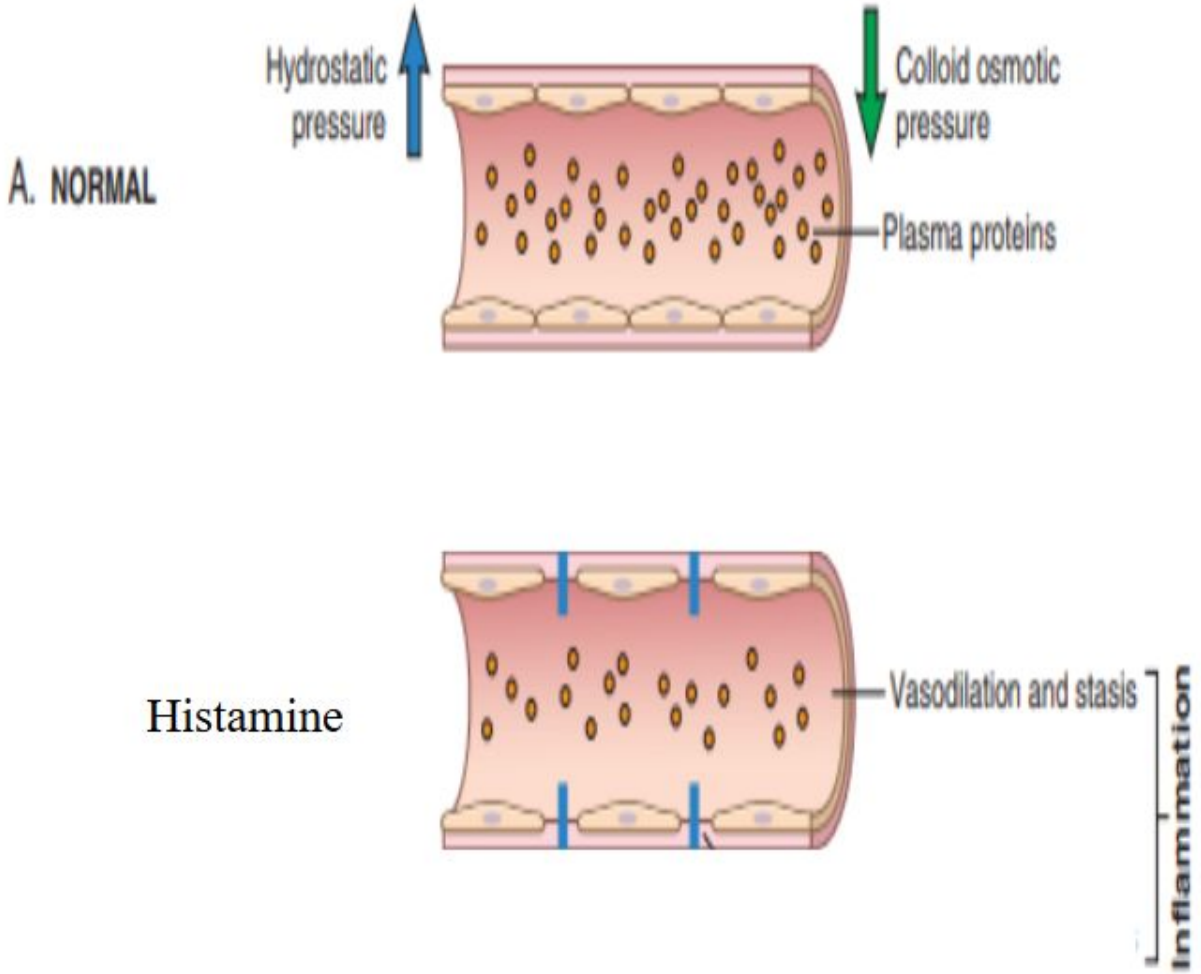
Acute Inflammation

Acute inflammation is a short-term process occurring in response to tissue injury, usually appearing within minutes or hours



Reactions of Blood Vessels in Acute Inflammation

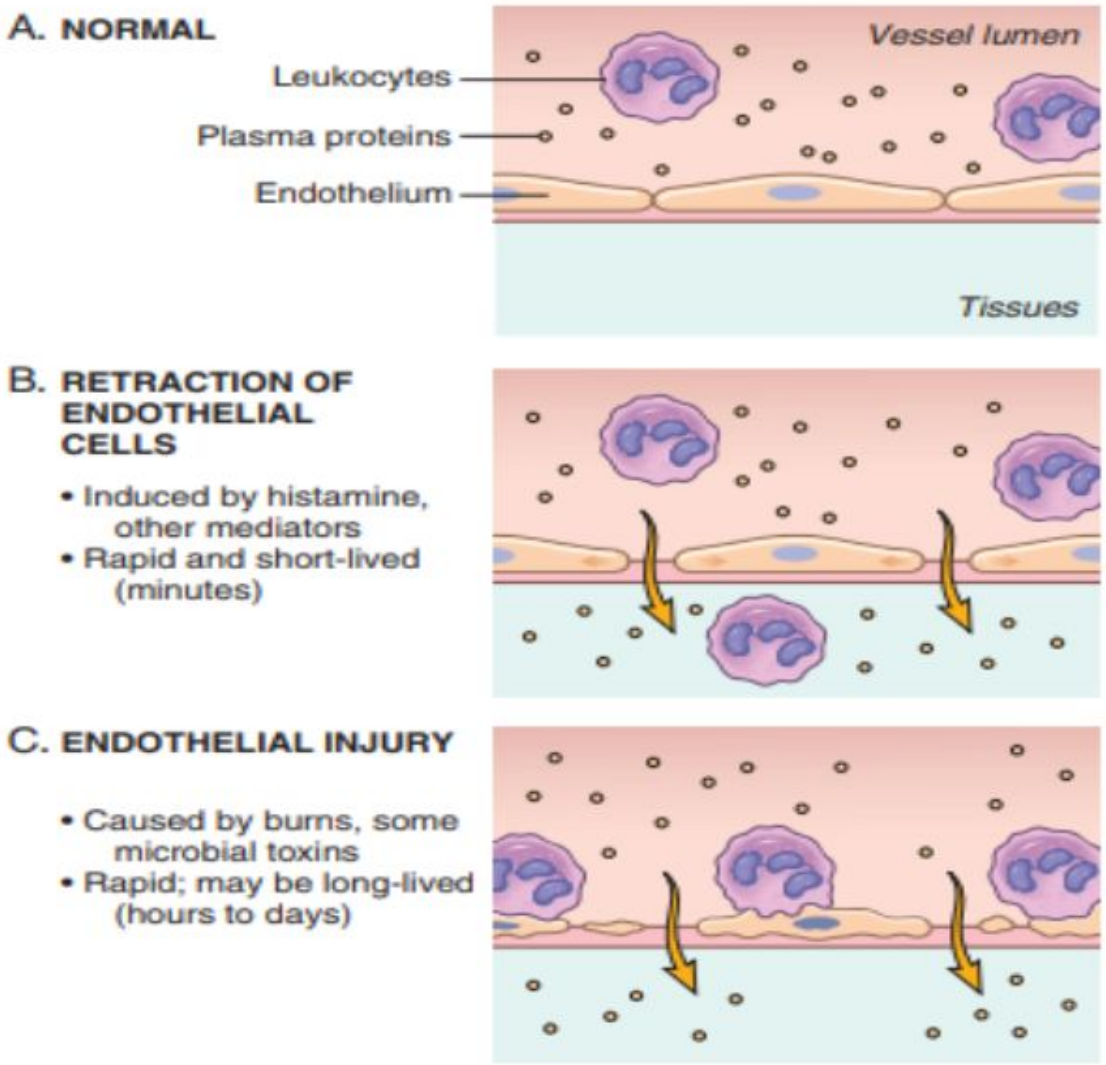
Changes in Vascular Flow and Caliber



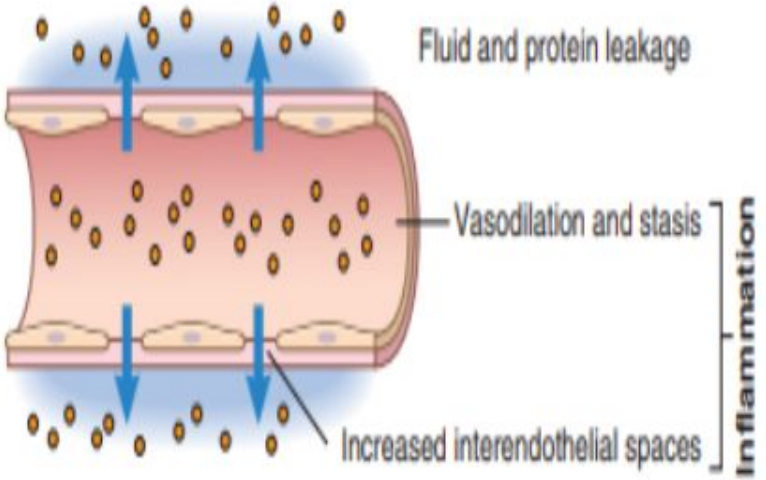


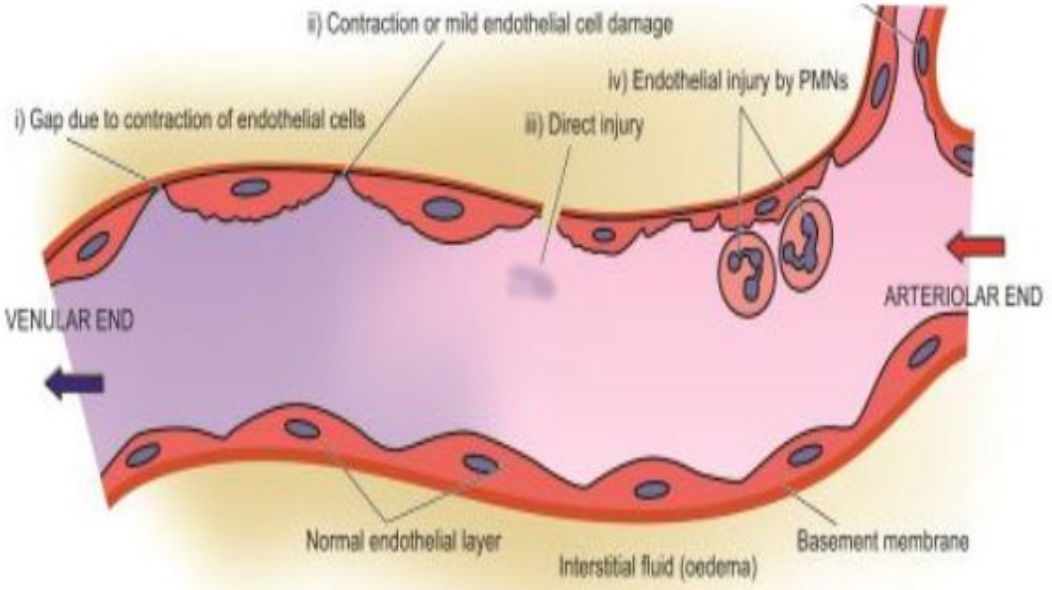
Reactions of Blood Vessels in Acute Inflammation

Increased Vascular Permeability (Vascular Leakage)

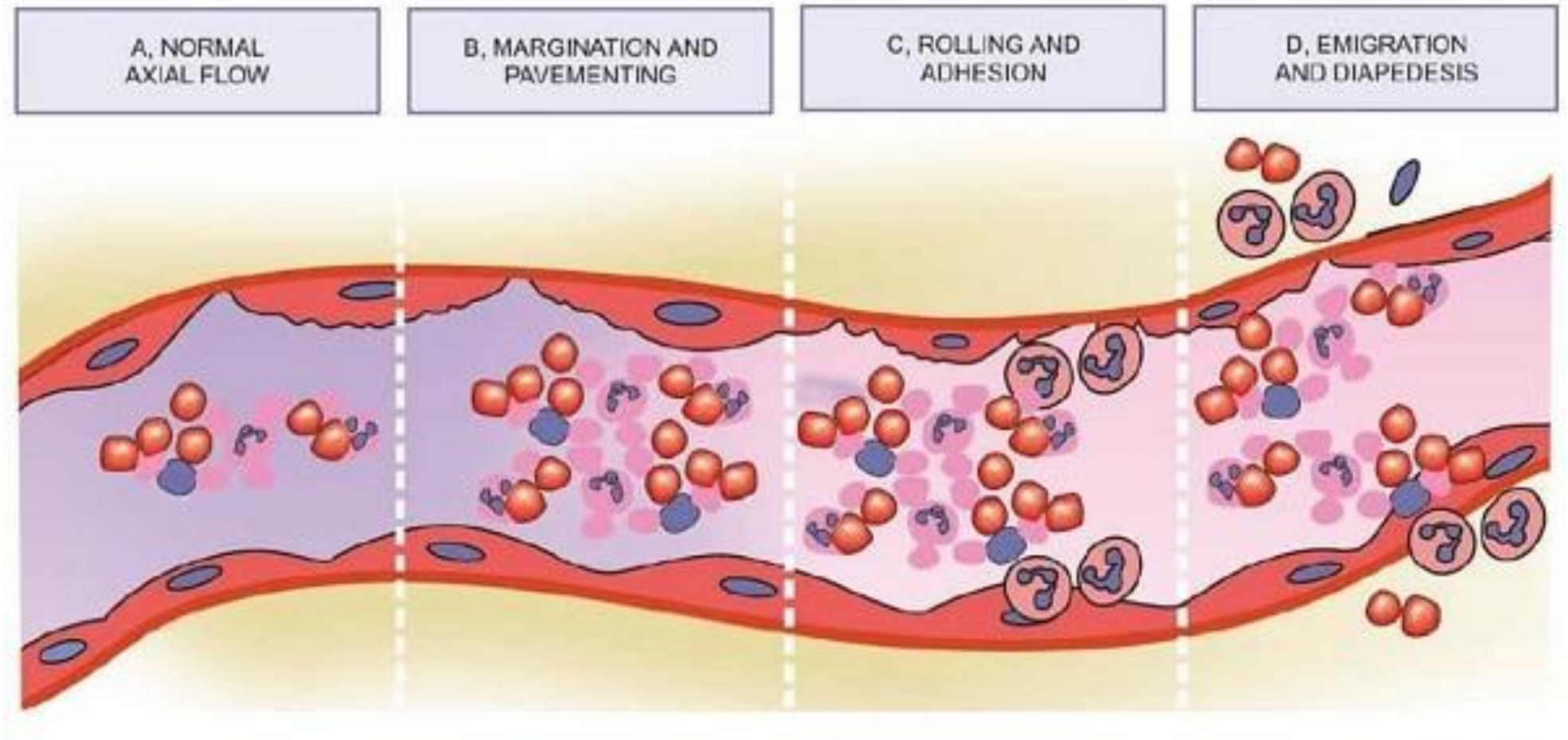


D **EXUDATE**
(high protein content, and may contain some white and red cells)

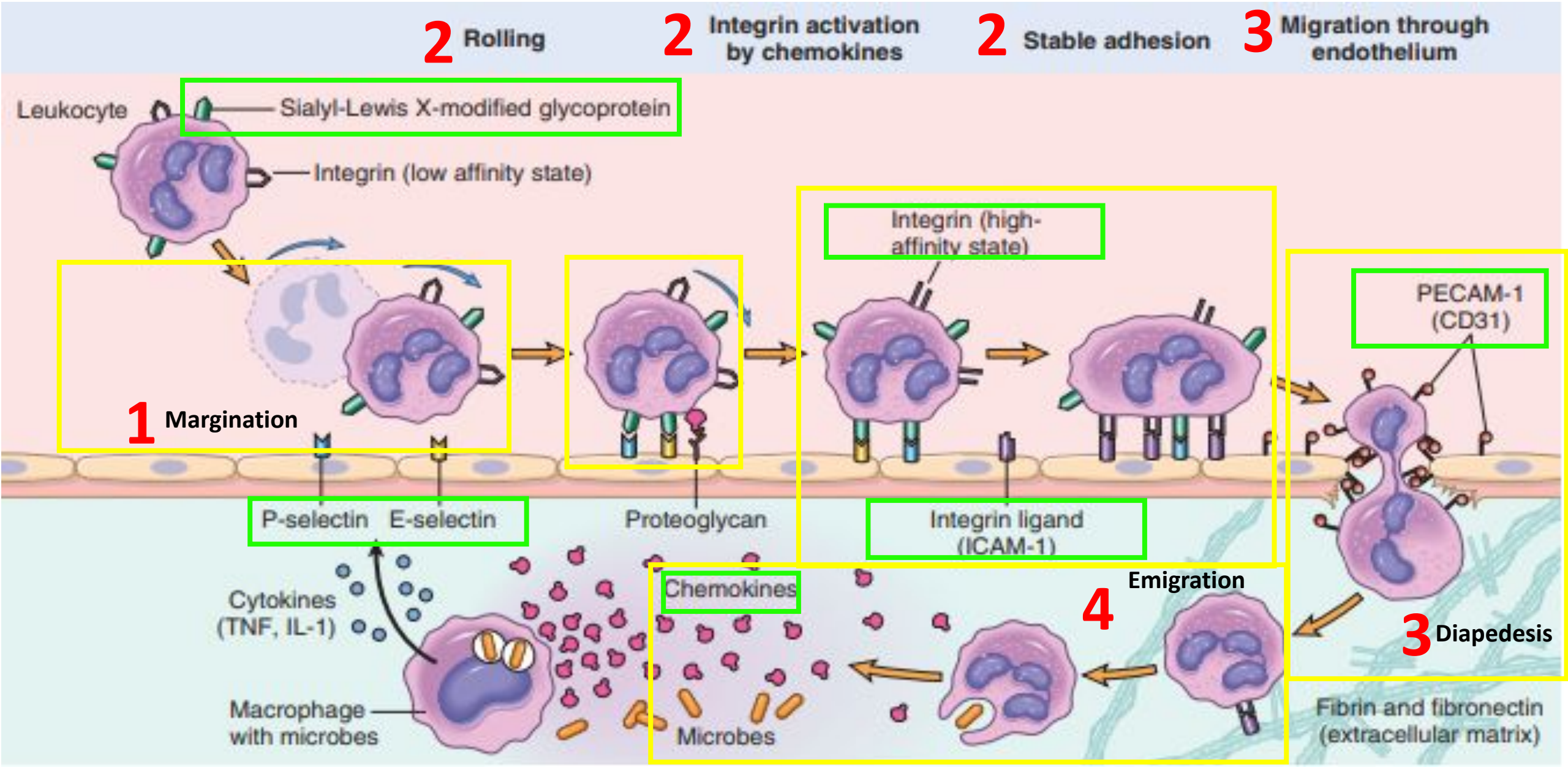




Leukocyte Recruitment to Sites of Inflammation

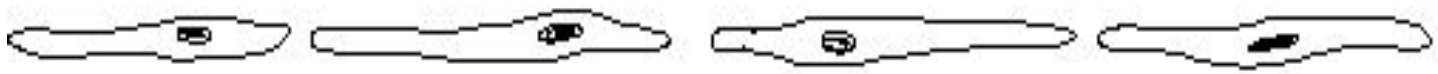


Leukocyte Recruitment to Sites of Inflammation



MARGINATION

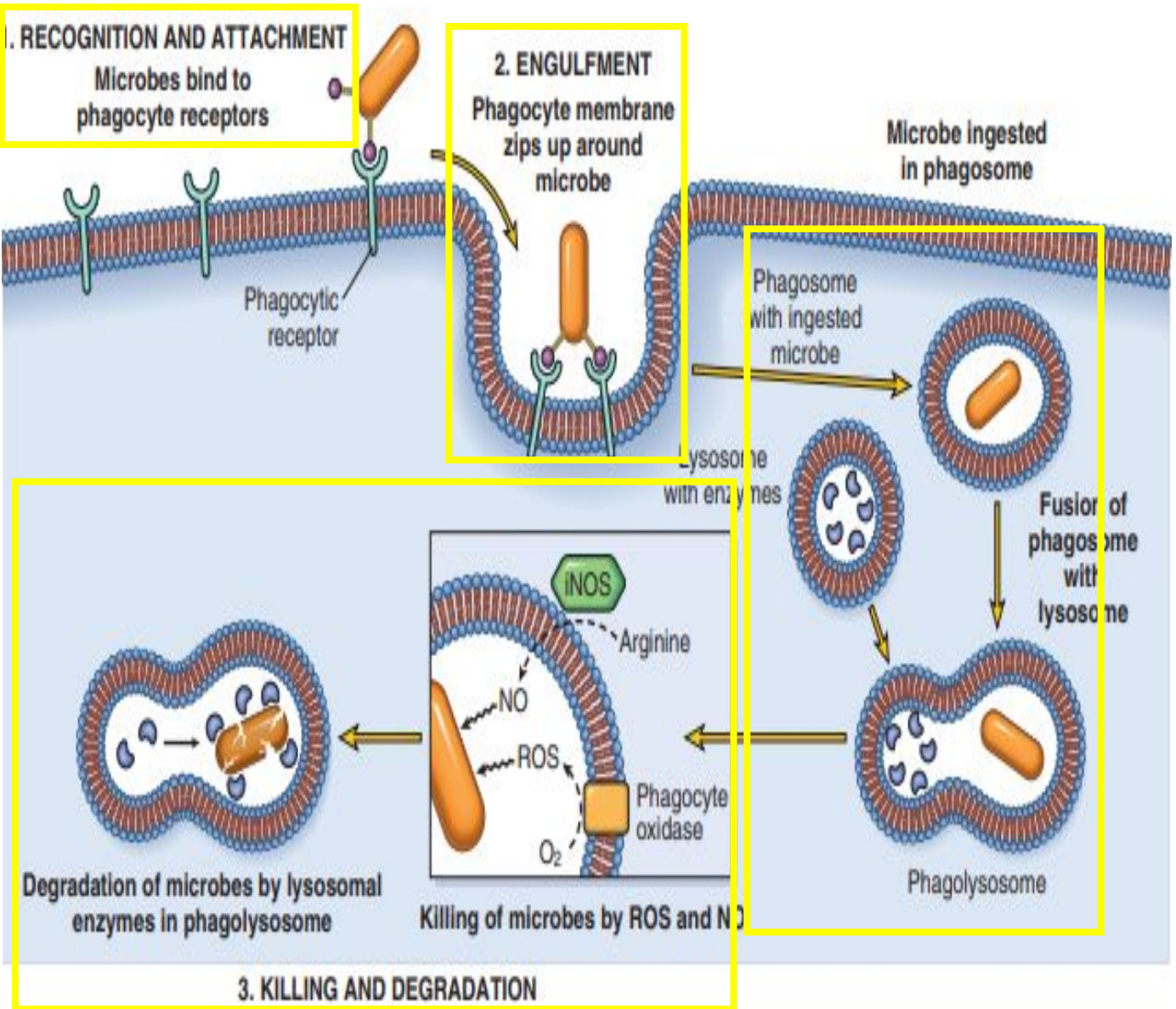


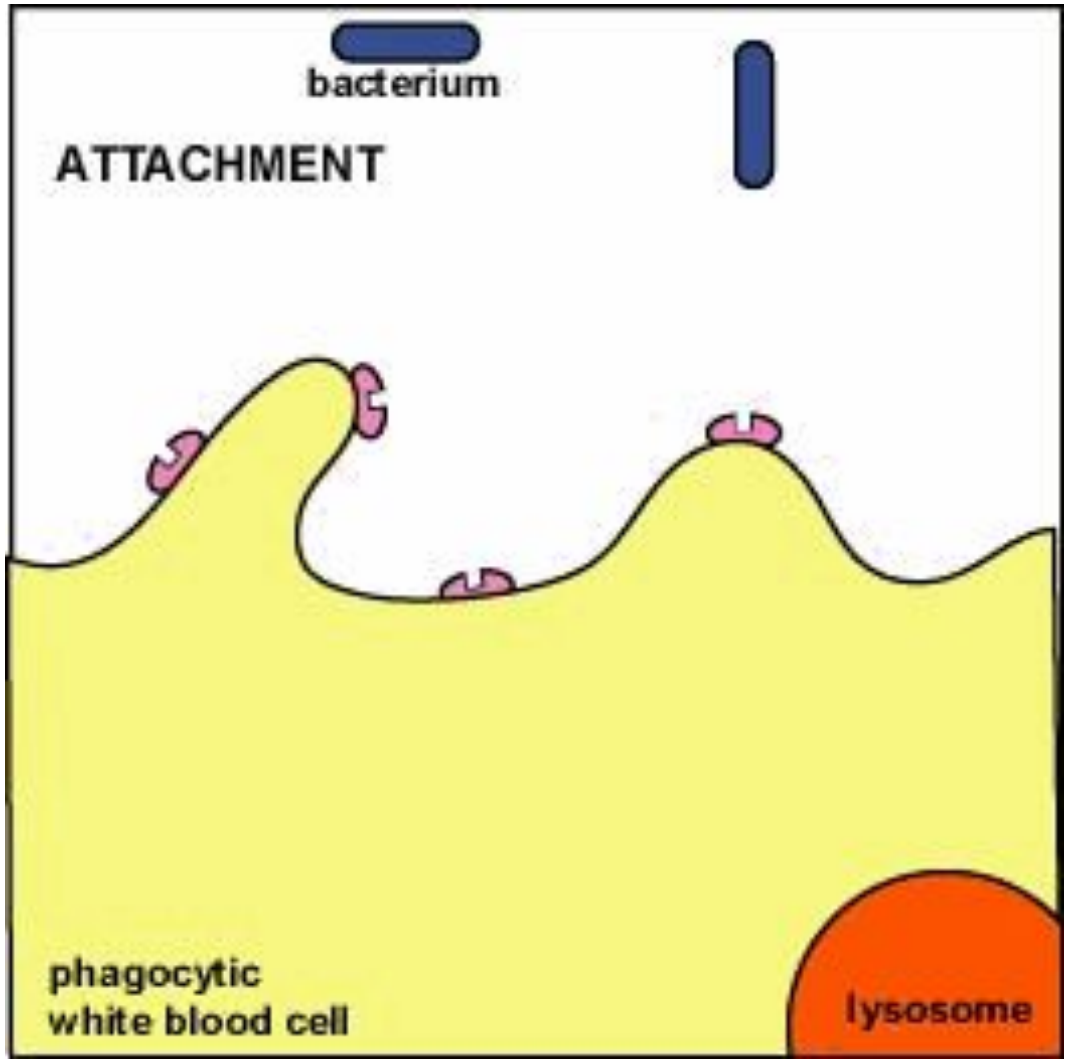




Phagocytosis and Clearance of the Offending Agent

Phagocytosis, Engulfment and Destruction





Phagocytosis and Clearance of the Offending Agent

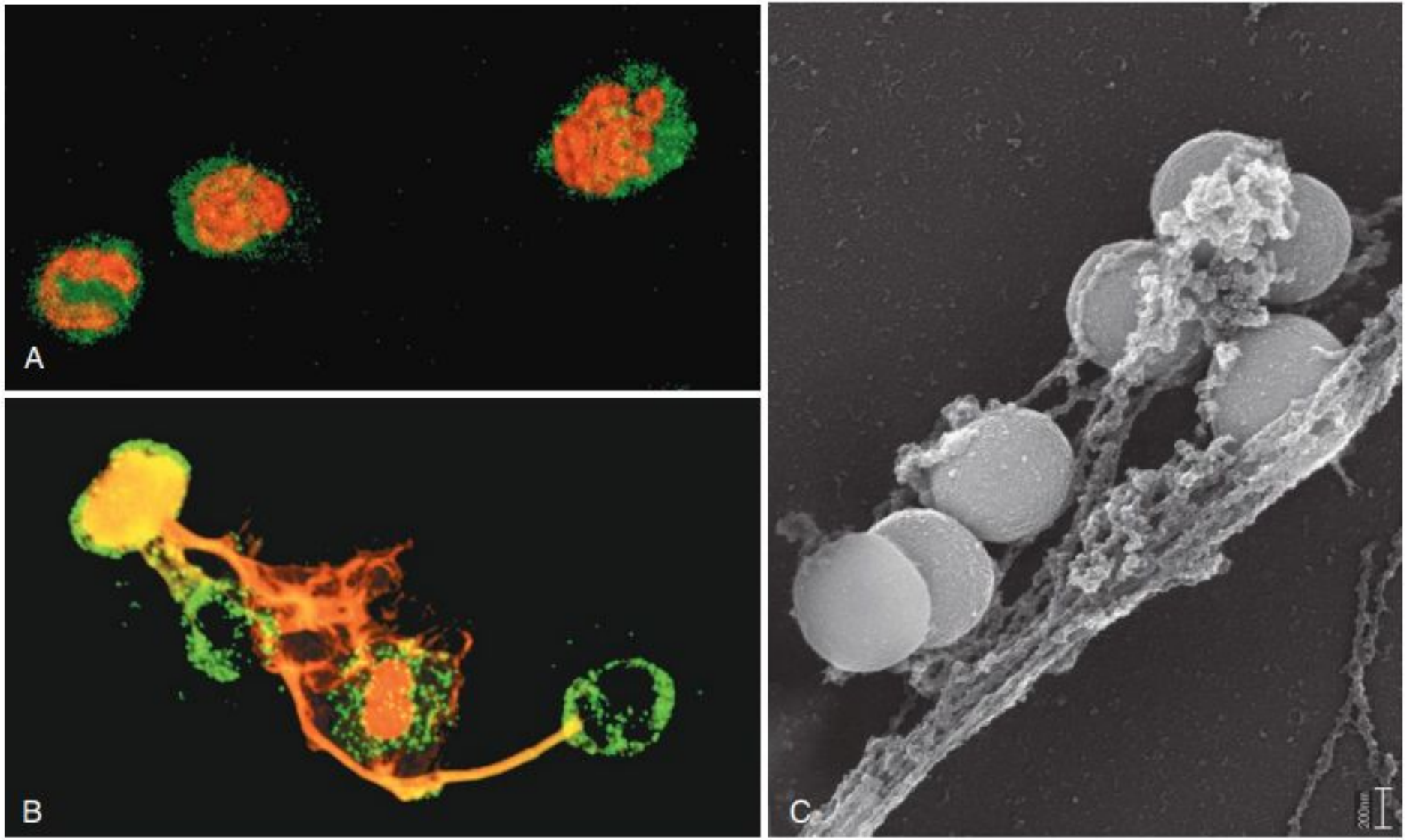
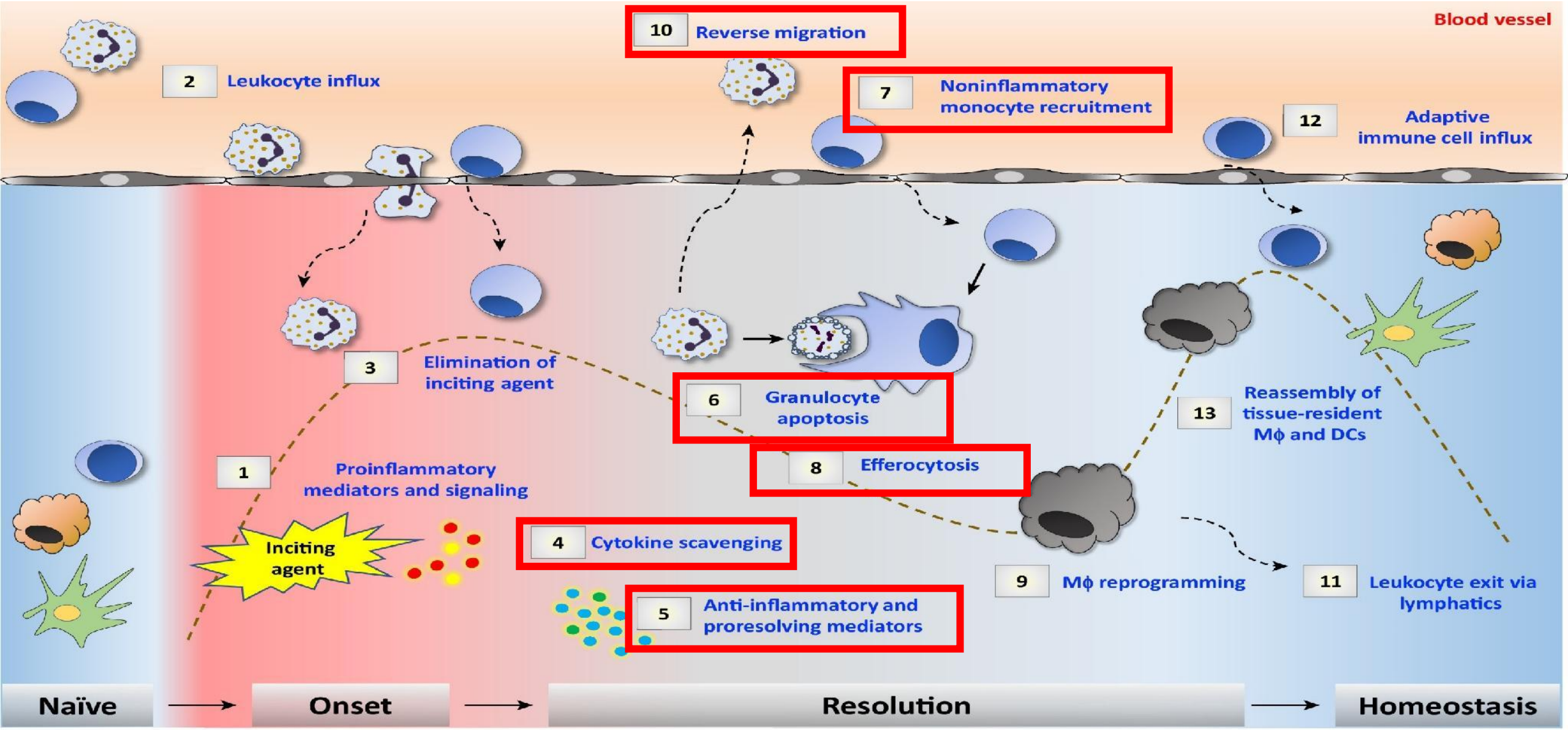


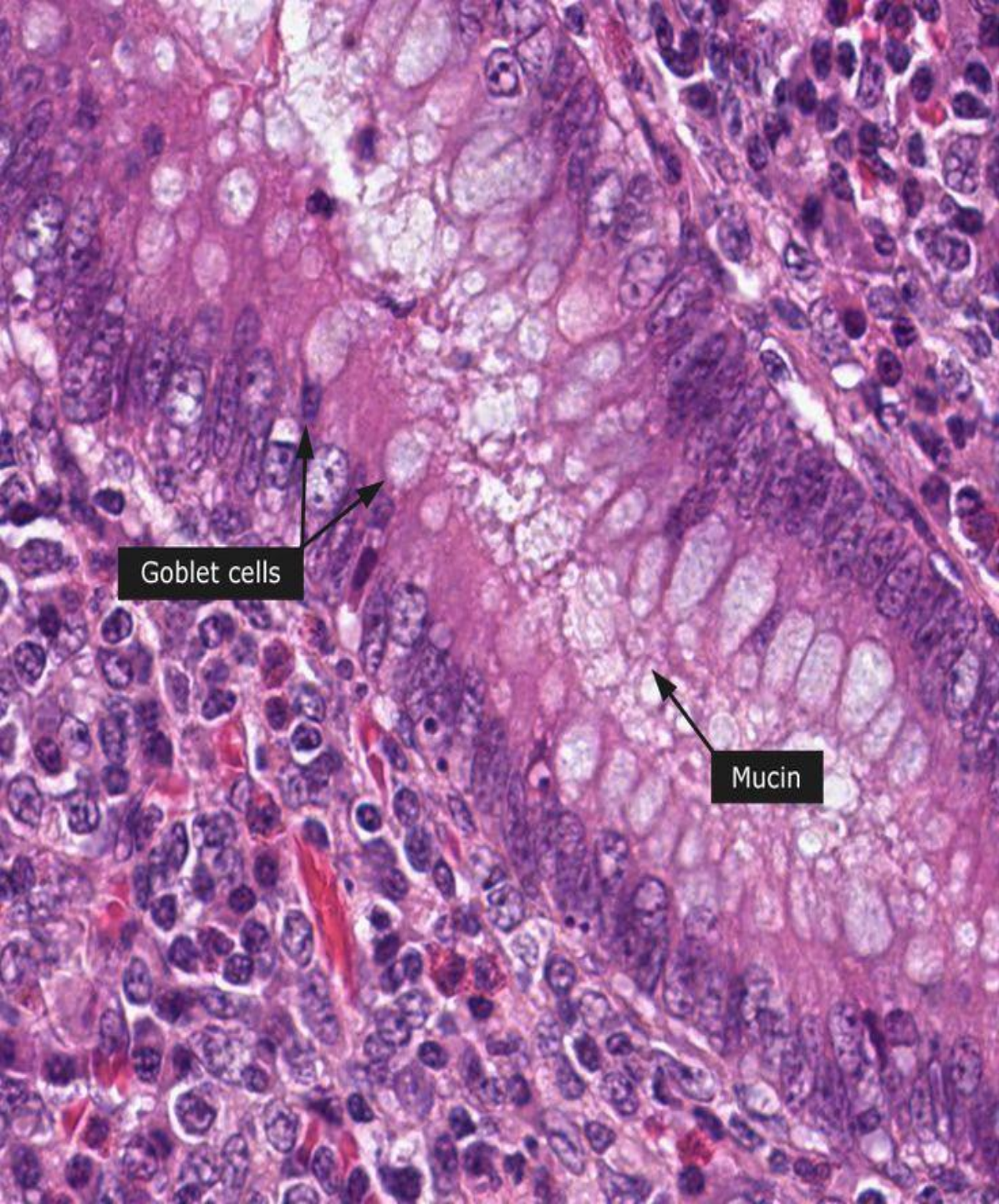
Figure 3-9 Neutrophil extracellular traps (NETs). **A,** Healthy neutrophils with nuclei stained red and cytoplasm green. **B,** Release of nuclear material from neutrophils (note that two have lost their nuclei), forming extracellular traps. **C,** An electron micrograph of bacteria (staphylococci) trapped in NETs. (From Brinkmann V, Zychlinsky A. Beneficial suicide: why neutrophils die to make NETs. Nat Rev Microbiol 2007;5:577, with permission.)

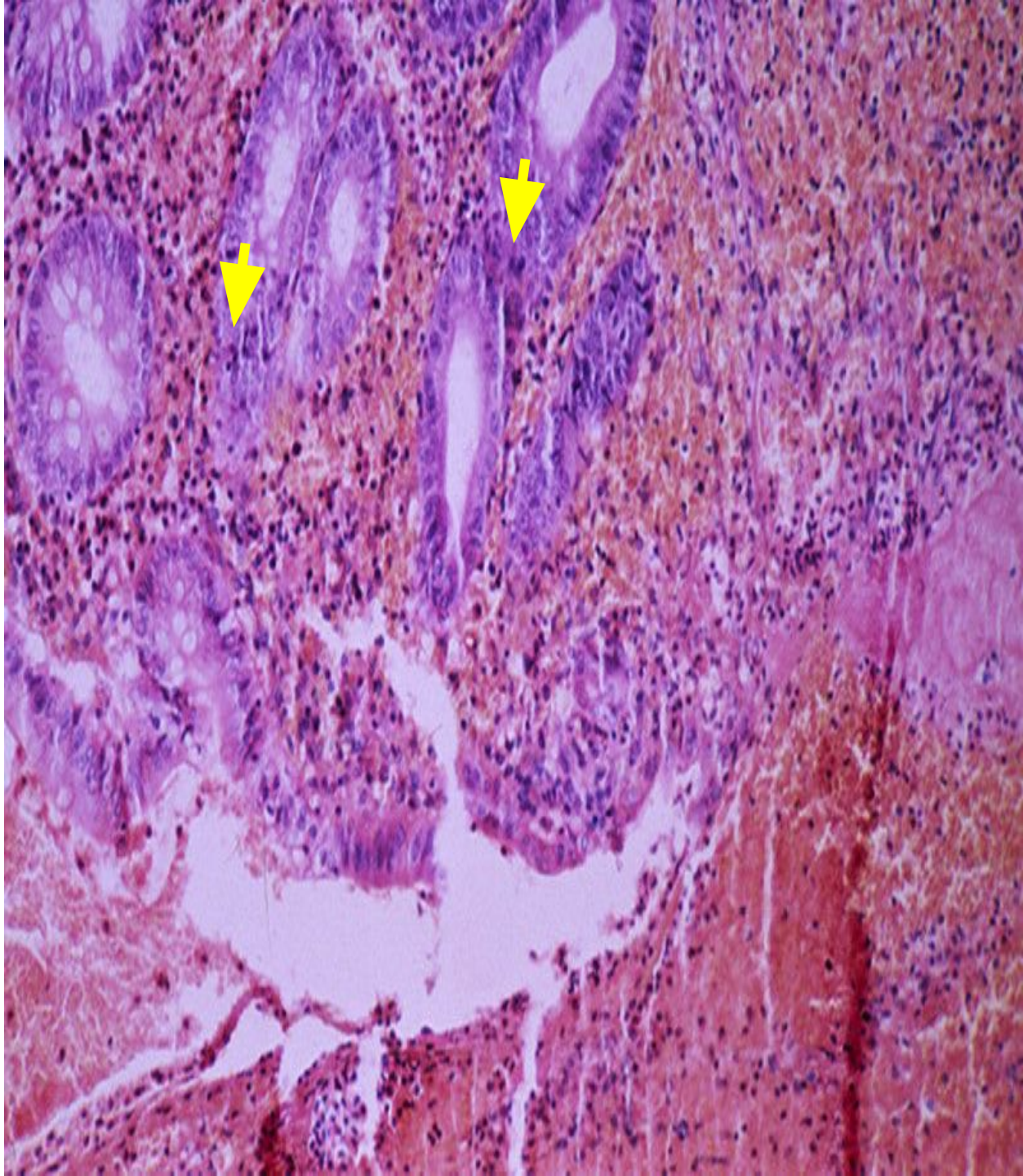
Activ

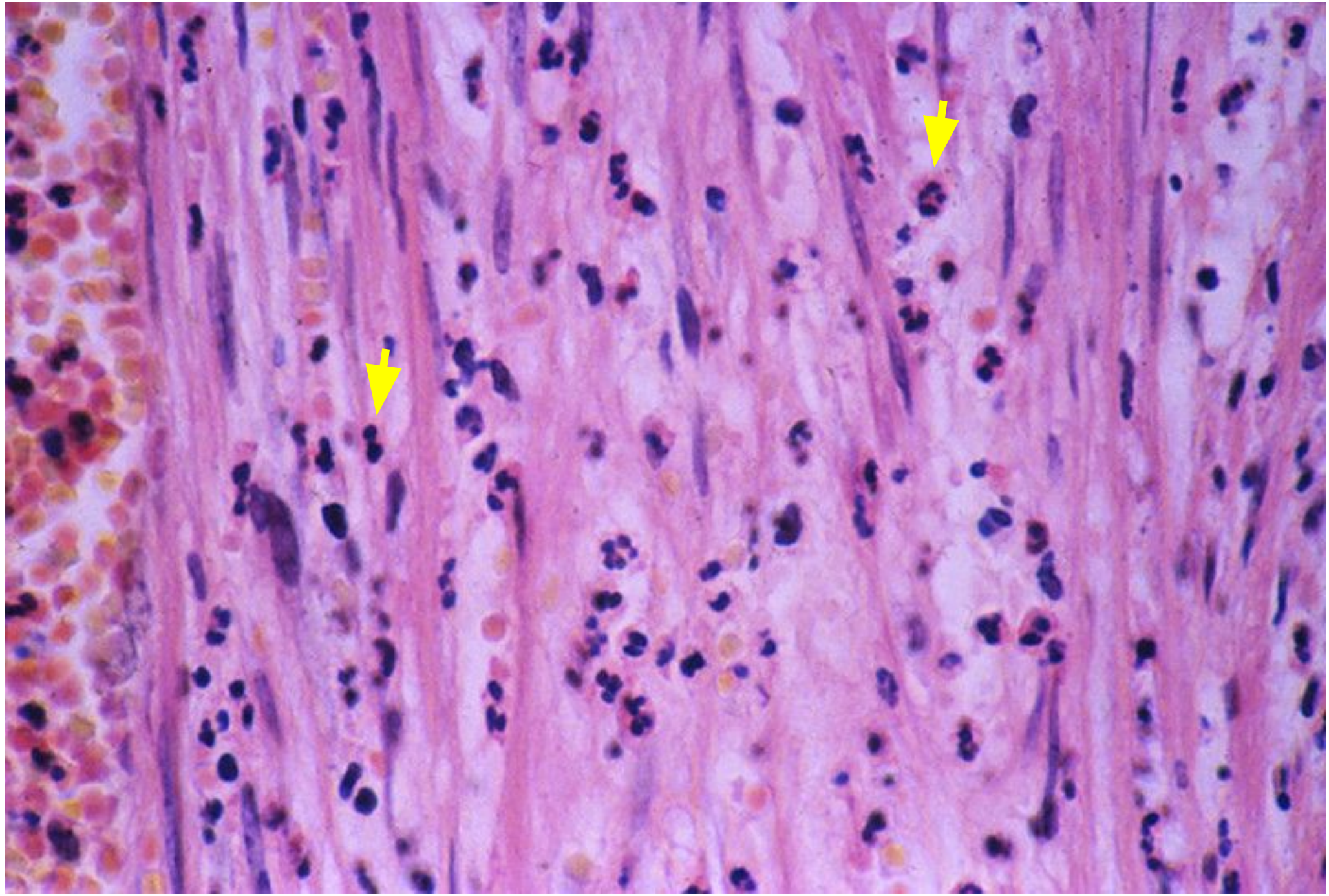
Termination of the Acute Inflammatory Response



An 18-year-old woman presented to the emergency department with a two-day history of nausea, vomiting and periumbilical and lower abdominal pain. Earlier that day, she experienced shaking chills and fever. Physical exam showed local tenderness and rigidity with rebound pain in the lower abdomen. Laboratory tests revealed a white blood cell count of 21,900/ dL. She underwent an exploratory laparotomy.

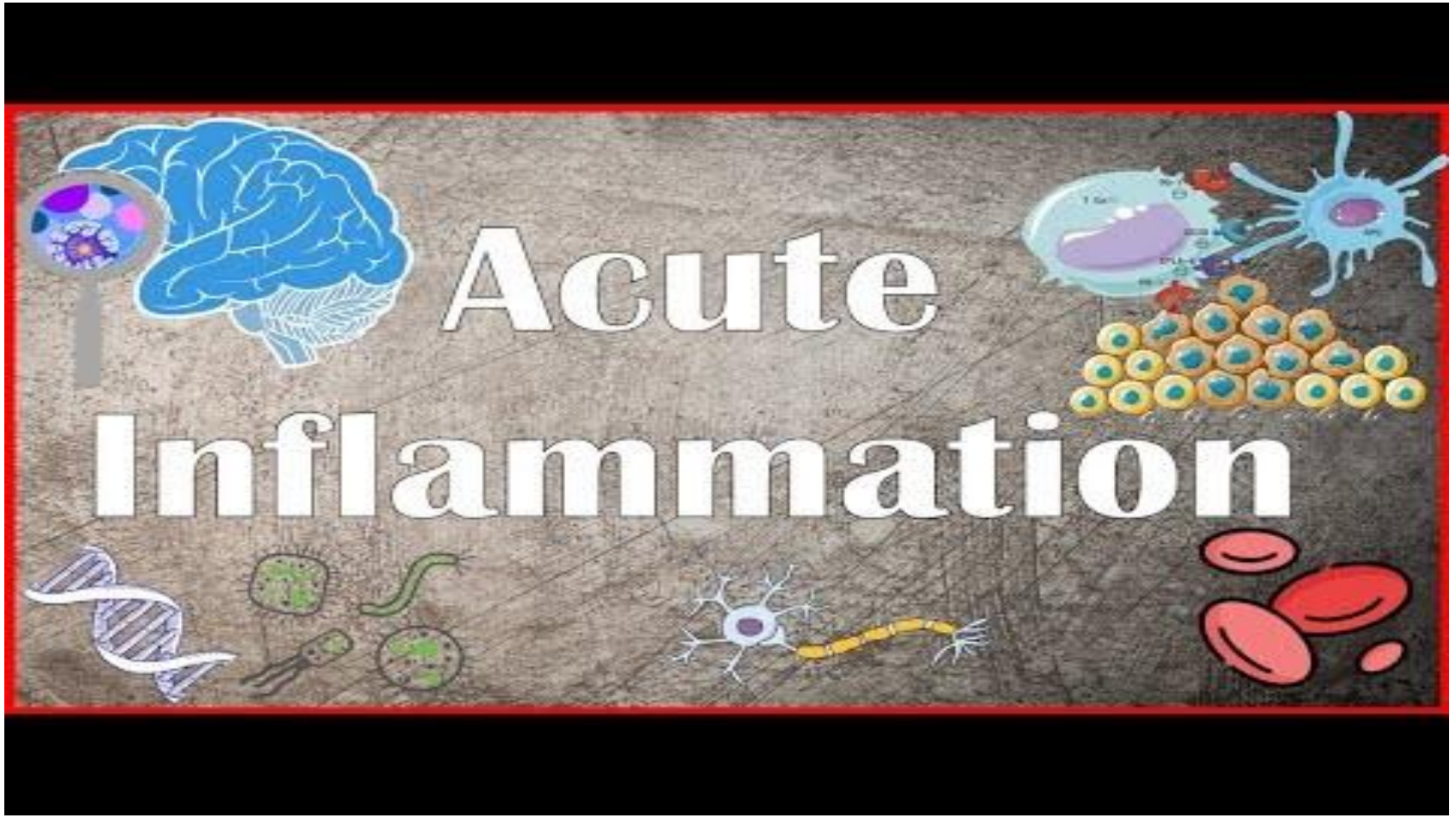

Goblet cells
Mucin





Additional Read






ME
38
INFLAMMATION


THANK YOU
THANK YOU
THANK YOU
THANK YOU

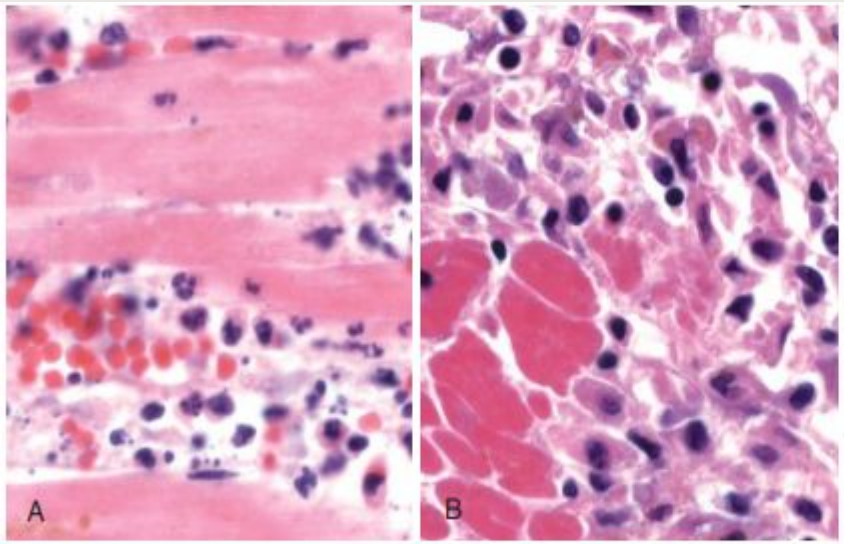
Disorders	Cells and Molecules Involved in Injury
Acute	
Acute respiratory distress syndrome	Neutrophils
Asthma	Eosinophils; IgE antibodies
Glomerulonephritis	Antibodies and complement; neutrophils, monocytes
Septic shock	Cytokines
Chronic	
Arthritis	Lymphocytes, macrophages; antibodies?
Asthma	Eosinophils; IgE antibodies
Atherosclerosis	Macrophages; lymphocytes
Pulmonary fibrosis	Macrophages; fibroblasts

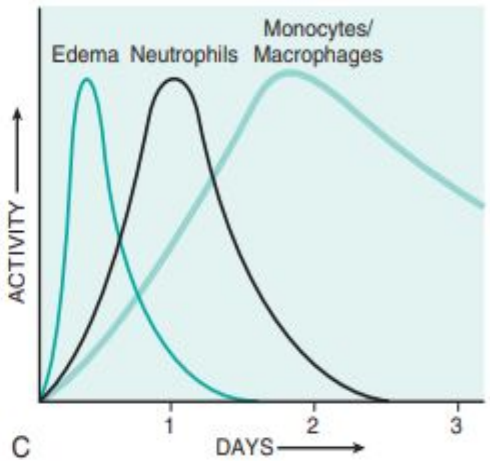
Feature	Acute	Chronic
Onset	Fast: minutes or hours	Slow: days
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury, fibrosis	Usually mild and self-limited	Often severe and progressive
Local and systemic signs	Prominent	Less

Edema

Connective tissue deposition and angiogenesis

Family	Molecule	Distribution	Ligand
Selectin	L-selectin (CD62L)	Neutrophils, monocytes T cells (naïve and central memory) B cells (naïve)	Sialyl-Lewis X/PNAd on GlyCAM-1, CD34, MAdCAM-1, others; expressed on endothelium (HEV)
	E-selectin (CD62E)	Endothelium activated by cytokines (TNF, IL-1)	Sialyl-Lewis X (e.g., CLA) on glycoproteins; expressed on neutrophils, monocytes, T cells (effector, memory)
	P-selectin (CD62P)	Endothelium activated by cytokines (TNF, IL-1), histamine, or thrombin	Sialyl-Lewis X on PSGL-1 and other glycoproteins; expressed on neutrophils, monocytes, T cells (effector, memory)
Integrin	LFA-1 (CD11aCD18)	Neutrophils, monocytes, T cells (naïve, effector, memory)	ICAM-1 (CD54), ICAM-2 (CD102); expressed on endothelium (upregulated on activated endothelium)
	MAC-1 (CD11bCD18)	Monocytes, DCs	ICAM-1 (CD54), ICAM-2 (CD102); expressed on endothelium (upregulated on activated endothelium)
	VLA-4 (CD49aCD29)	Monocytes T cells (naïve, effector, memory)	VCAM-1 (CD106); expressed on endothelium (upregulated on activated endothelium)
	α4β7 (CD49DCD29)	Monocytes T cells (gut homing naïve effector, memory)	VCAM-1 (CD106), MAdCAM-1; expressed on endothelium in gut and gut-associated lymphoid tissues
Ig	CD31	Endothelial cells, leukocytes	CD31 (homotypic interaction)





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